

Factory Method Pattern

The Factory Method pattern defines an interface for creating an object, but lets subclasses decide which class to instantiate. Rather than calling a constructor directly, client code calls a factory method, and the specific subclass is decided by which factory subclass is used at runtime.

When to use it

Use Factory Method when a class cannot anticipate the class of objects it must create, or when a class wants its subclasses to specify the objects it creates. It is especially useful when object creation involves logic that would otherwise be duplicated across many call sites in client code.

Abstract Factory vs Factory Method

Abstract Factory is Factory Method's big sibling: instead of one factory method producing one kind of product, an Abstract Factory groups several related factory methods together to produce a whole family of related objects, guaranteeing that objects produced together are compatible with each other, for example a UI toolkit producing a matching button, checkbox, and scrollbar for a given operating system's look and feel.

Singleton Pattern

Singleton restricts a class to a single instance and provides a global access point to it. It is one of the most overused patterns in practice because it is frequently reached for as a substitute for proper dependency injection, which makes code harder to test since a singleton's global, shared state can leak between otherwise-independent unit tests.